

Coupled Diffusion Posterior Sampling for Unsupervised Hyperspectral and Multispectral Images Fusion

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Abstract—Hyperspectral images (HSIs) and multispectral images (MSIs) fusion is a hot topic in the remote sensing society. A high-resolution HSI (HR-HSI) can be obtained by fusing a low-resolution HSI (LR-HSI) and a high-resolution MSI (HR-MSI) or RGB image. However, most deep learning-based methods require a large amount of HR-HSIs for supervised training, which is very rare in practice. In this paper, we propose a coupled diffusion posterior sampling (CDPS) method for HSI and MSI fusion in which the HR-HSIs are no longer required in the training process. Because the LR-HSI contains the spectral information and HR-MSI contains the spatial information of the captured scene, we design an unsupervised strategy that learns the required diffusion priors directly and solely from the input test image pair (the LR-HSI and HR-MSI themselves). Then, a coupled diffusion posterior sampling method is proposed to introduce the two priors in the diffusion posterior sampling which leverages the observed LR-HSI and HR-MSI as fidelity terms. Experimental results demonstrate that the proposed method outperforms other state-of-the-art unsupervised HSI and MSI fusion methods. Additionally, this method utilizes smaller networks that are simpler and easier to train without other data.

Index Terms—Hyperspectral images, fusion, diffusion posterior sampling, unsupervised learning.

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) contain hundreds or thousands of continuous spectra and can present the spectral curve of the earth observations. With these advantages, HSIs have been widely used in many applications, such as land-cover mapping, target detection, anomaly detection, and change detection [1], [2]. However, in the HSI imaging system,

the amount of incident energy is limited and the high spectral resolution results in a low spatial resolution of HSI [3], [4]. In addition, improving the spatial resolution of HSI in the imaging system is difficult and costly due to hardware limitations. Therefore, enhancing the spatial resolution of HSI can be achieved through digital image processing methods [5], [6], [7].

The most commonly used way to HSI super-resolution is to fuse a high-resolution multispectral image (HR-MSI) and a low-resolution HSI (LR-HSI) capturing the same scene [8], [9], [10], [11], [12]. The LR-HSI and HR-MSI are easily obtained in the imaging system, thus fusing the two images will always achieve promising HR-HSI.

The existing HSI and MSI fusion methods can be roughly divided into 3 categories, detail-injection-based methods, variational optimization-based methods, and deep learning-based methods. In the detail-injection-based methods, the spatial details are extracted from the high-resolution images and injected into the fused images. Although these methods are fast and computationally efficient, the fused images always exhibit non-negligible spectral distortion. In variational optimization-based methods, the HR-HSI is first described with a matrix or tensor factorization model. Then, each component in the variational model is optimized using priors as regularization terms. Variational optimization-based methods require the degradation parameters to be known in prior, and the fusion quality heavily relies on the handcrafted priors. With the fast development of deep learning (DL), DL-based methods have attracted more attention in HSI and MSI fusion. In the DL methods, a mapping from the LR-HSI and HR-MSI to the HR-HSI is learned from the training pairs with a human-designed deep convolutional neural network (CNN) [13], [14], [15], [16], [17] or the Transformer network [18], [19], [20], [21]. Although the DL-based methods achieved superior results, the training process requires a large number of HR-HSI and LR-HSI, HR-MSI pairs, which is unrealistic in real applications. To address this problem, some unsupervised methods are proposed that leverage the autoencoder networks to fuse the spatial information in HR-MSI and spectral information in LR-HSI. In addition, the degradation parameters, including the spectral response function (SRF) and point spread function (PSF) can also be learned from the networks. However, these

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methods are fully data-driven, and can seldom incorporate handcrafted priors.

In this paper, we propose a coupled diffusion posterior sampling (CDPS) method for HSI and MSI fusion, which learns priors directly from the observations rather than relying on other training data. In the proposed method, the fusion problem is taken as an inverse problem and the input data, LR-HSI and HR-MSI, are the observations. We design a diffusion posterior sampling method to generate the HR-HSI from the pure noise with the given observation. However, in the diffusion posterior sampling, a diffusion model for the HR-HSI should be trained with a large number of the HR-HSIs captured by the same sensor. In fact, the diffusion model can learn the distribution of the training data and samples from the learned distribution. The higher the dimension of the data, the harder the training process. However, the HR-HSI always exhibits higher dimensionality than traditional natural images. Therefore, training a diffusion model directly for HR-HSI is very challenging due to the high-dimensional signal and the limited training data.

To address these problems, we propose to decompose the HR-HSI into two low-dimensional components with the subspace model. One is the spatial representation of abundance images and the other is the spectral basis or the spectral endmembers. The abundance images reveal the spatial distribution of each endmember, corresponding to the spatial structure of the HR-MSI. Therefore, we train a spatial diffusion model from the spatial information contained in the HR-MSI. Endmembers consist of multiple components representing different ground materials. To learn the distribution of the endmember, we use real spectra which are contained in the LR-HSI. Therefore, we learn a spectral diffusion model from the spectral information contained in the LR-HSI. With the learned spatial and spectral diffusion models, we generate an HR-HSI in the posterior sampling by coupling the two diffusion models. The resulting HR-HSI also adheres to the degradation process of the LR-HSI and HR-MSI.

In summary, the highlights of the proposed coupled diffusion posterior sampling (CDPS) method can be concluded in three folds.

1. We propose the posterior sampling framework specifically designed for HR-HSI reconstruction. This core innovation explicitly integrates the physical degradation models (spatial blurring and spectral downsampling) directly into the sampling process. By conditioning on the observed LR-HSI and HR-MSI, our method generates HR-HSI samples that inherently adhere to the physics of the imaging system while respecting the learned prior distribution of the observed hyperspectral scenes. This fundamentally addresses the ill-posedness and ambiguity inherent in fusion.
2. To overcome the prohibitive cost of training a full-dimensional HR-HSI diffusion model, we introduce a novel subspace-coupled framework. By leveraging the spectral low-rank property of HSIs, we decompose the problem into two low-dimensional, easily trainable diffusion models: one capturing the spatial prior (via the HR-MSI subspace) and one capturing the spectral

prior (via the LR-HSI subspace). This unique coupling makes high-resolution posterior sampling computationally tractable.

3. We introduce a novel unsupervised framework that eliminates the requirement for a large, externally trained model or a paired dataset. By leveraging the subspace decomposition, the spatial and spectral diffusion models are trained directly and solely from the single test image pair (the HR-MSI and LR-HSI) prior to sampling. This training-data-free strategy, facilitated by our coupled design, significantly enhances the practical applicability of the method in scenarios where collecting paired training data is infeasible.

The remainder of the paper is organized as follows. Section II describes the related work. In Section III, the proposed CDPS method for HSI fusion is introduced in detail. Section IV conducts extensive experiments on HSI data sets to demonstrate the advantages of the proposed method. Finally, conclusions are drawn in Section V.

II. RELATED WORKS

A. Diffusion Model for Inverse Problem

Recently, generative models have achieved great attention in computer vision and multi-modal generation. The diffusion model, which learns the potential distribution of the data, presents superior performance in image generation. It first learns the problem-agnostic diffusion models pre-trained with a large amount of images. With the pre-trained model, the images can be recovered in a plug-and-play way which also combines the measurement process. The denoising diffusion restoration model (DDRM) [22] was first proposed which uses the pre-trained diffusion model as the prior motivated by variational inference. To learn a better likelihood score, [23] projected the image onto the measurement subspace. However, it cannot handle the measurement with noise. Denoising Diffusion Null-Space Model (DDNM) [24] was proposed which can handle arbitrary linear inverse problems. Addressing the key challenge of modeling the likelihood in diffusion-based inverse methods, Diffusion Posterior Sampling (DPS) [25] leveraged the Laplace approximation to estimate the distribution of the original data. Further, a parallel diffusion model was proposed [25] which can process multiple variables via the parallel reverse diffusion process. In [26], a self-supervised diffusion Denoising Diffusion Spatial-Spectral Model (DDS2M) was proposed to solve the problem of HSI restoration. It restores the HSI by generating the spectral and spatial subspace in the diffusion sampling. Although it achieves promising results in HSI restoration, the networks are randomly initialized. Therefore, the lack of prior distribution for the HSI restricts the applicability of the method. In [27], an unsupervised HSI pansharpening method was proposed via the low-rank diffusion model. The low-rank representation parameters are generated by the pre-trained diffusion model. However, the pre-trained model is obtained by natural images which may not be suitable for HSI and the rank is fixed according to the natural images. In [28], a novel diffusion model with disentangled modulations is proposed for HSI and MSI images fusion which introduces

two modulation modules to disentangle style and frequency parts from diverse domains. But the pre-trained model cannot be used for sampling with different datasets.

In fact, the advantage of the diffusion model lies in the well-trained networks which can reveal the underlying distribution of the data. However, the data used in the training process is very large due to the high-dimensional signal. This is not a practical way in real applications, and a more convenient way or easily trained networks should be leveraged.

B. HSI and MSI Fusion

HSI has abundant spectral information, but the spatial resolution is not satisfying. To increase the spatial resolution of HSI, high-resolution remote images (HR-MSI, panchromatic, RGB *et. al*) can be leveraged to provide more spatial information which is called fusion-based super-resolution of HSI. In this paper, we use the HSI and MSI fusion to represent the super-resolution combining LR-HSI and HR-MSI/panchromatic/RGB image. The HSI and MSI fusion methods can be roughly divided into three categories – detail-injection-based methods, variational optimization-based methods and DL-based methods. In detail-injection-based methods, one component of HSI is substituted by the high-resolution images. For instance, intensity-hue-saturation (IHS) [29], [30] replaces the intensity channel of the low-resolution image in the domains of intensity, hue, and saturation with a panchromatic (PAN) image. Similarly, the Principal Component Analysis (PCA) approach [31] replaces the first principal component of the low-resolution HSI with a high spatial resolution image. However, it's important to note that while the detail-injection-based approaches are effective in enhancing spatial resolution, they often introduce spectral distortions in the reconstructed high-resolution HSI. Variational optimization-based methods usually build a model to characterize the HR-HSI and solve the unknown variables in the model with the observed LR-HSI and HR-MSI. Yokoya et al. [32] first built the coupled non-negative matrix factorization (CNMF) model to fuse the LR-HSI and HR-MSI. It uses the linear unmixing model to decompose the HR-HSI into the endmembers and abundance maps and solve the two parts one by one. To characterize the sparseness of the HR-HSI, Dong et al. [33] proposed the non-negative structured sparse representation fusion method which estimates the spectral dictionaries and sparse coding coefficients simultaneously. Some researchers introduce HSI's priors into the fusion model. Simoes et al. [34] proposed the vector-total variation (VTV) regularization to fuse HR-HSI. In [35], the group spectral embedding and low-rank prior were introduced to the HR-HSI. In addition, tensor-based model is developing rapidly because it can well preserve the multi-dimensional structure of HR-HSI. Dian et al. [36] proposed the Tucker decomposition-based fusion model incorporating the non-local dictionary, spectral dictionary, and two spatial dictionaries. Further, the authors proposed the coupled Tucker factorization model (CSTF) [37] by modeling the HR-HSI to be estimated as one third-order tensor. In [3], a CP decomposition-based fusion method was proposed by grouping the similar 3D patches in HR-HSI and

introducing the low-rank constraint. To preserve the high-order correlation of HR-HSI, the high-order couple tensor ring model (HCTR) [38] was proposed by rearranging the HR-HSI as the high-order tensor.

The DL-based methods usually use the observed data to learn the mapping from LR-HSI to HR-HSI. Yang et al. [39] proposed the dual branch CNN to extract the information separately and fuse the features later. To preserve the spectral structure, Zhang et al. [40] proposed a physical model which incorporates the spatial edge and spectral edge loss. In [15], the well-trained CNN denoiser was leveraged to achieve high-performance super-resolution. In [41], Xie et al. proposed a novel MHFNet model that introduces the LR-HSI and HR-MSI imaging processes by building an unfolding end-to-end deep architecture. To introduce more prior information on the HSI, a deep image prior learned from unsupervised training was imposed in a deep blind HSI super-resolution model [42]. However, all of these methods need to train a specific deep-learning model according to different observations. Thus, the generalization of these methods is limited. In [43], Zhang et al. developed a two-stage fusion network including a supervised fusion module and an adaptation degeneration network. In this work, the unknown degeneration can be learned in an unsupervised learning scheme on the observed images. Recently, diffusion model was also used to fuse HSI and MSI. In [44], a dual conditional diffusion model-based fusion network (DCDMF) was proposed which uses two noise prediction networks with different conditional inputs. In [27], a low-rank diffusion model was proposed for HSI pansharpening in which a pre-trained deep diffusion model is leveraged to introduce the image prior under the Bayesian rule. In [45], a spatial-spectral-bilateral cycle-diffusion framework (S²CycleDiff) method was proposed which step-wise generates the HR-HSI by learning a customized conditional cycle-diffusion network. However, these diffusion model-based methods require training a large diffusion model in prior or using the pre-trained image model in the sampling process. The training of the large model is time-consuming and requires a large number of high-quality data which is not always feasible in real applications. To overcome these issues, we propose the CDPS method which greatly reduces the training of the model and requires no other data for training. Thus, the method can be widely used.

III. PROPOSED METHOD

In this section, we will present the proposed CDPS method in detail. The proposed method includes the training process and the diffusion posterior sampling process.

A. Problem Formulation

The observation of the HSI fusion is a pair of images, including an LR-HSI $\mathcal{X} \in \mathbb{R}^{w \times h \times B}$ and an HR-MSI $\mathcal{Y} \in \mathbb{R}^{W \times H \times b}$, where w and h represent the width and height of the LR-HSI, B represents the band number of the LR-HSI, W and H represent the width and the height of the HR-MSI, b represents the band number of HR-MSI. Here, we assume that $W > w$, $H > h$, and $B > b$. The goal of HSI fusion is to reconstruct the HR-HSI $\mathcal{Z} \in \mathbb{R}^{W \times H \times B}$ by fusing the LR-HSI and HR-MSI.

In general, the degradation model for the HSI fusion can be formulated as the following functions

$$\mathcal{X} = \mathcal{Z} \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2 + \mathcal{N}_X, \quad (1)$$

$$\mathcal{Y} = \mathcal{Z} \times_3 \mathbf{S} + \mathcal{N}_Y, \quad (2)$$

where $\mathbf{P}_1 \in \mathbb{R}^{w \times w}$ and $\mathbf{P}_2 \in \mathbb{R}^{h \times h}$ are the PSF matrices, and $\mathbf{S} \in \mathbb{R}^{b \times b}$ represents the SRF matrix, $\times_n$ denotes the mode- n tensor-matrix product, $\mathcal{N}_X$ and $\mathcal{N}_Y$ are the measurements noise, which is assumed to be the Gaussian noise. Although we have two observations in the HSI fusion problem, the HR-HSI to be estimated still has a higher dimensionality than the observations. Therefore, HSI and MSI fusion represent an inverse problem with multiple solutions that conform to the degradation models without the need for additional priors or data.

B. Diffusion Posterior Sampling for HSI and MSI Fusion

Diffusion models are generative frameworks that learn data distributions through a *noising process* and its reverse. Formally, given data $\mathbf{x}_0 \sim p_{\text{data}}$. There are both forward process and reverse process in the diffusion model. In the forward process, data are corrupted via a stochastic differential equation (SDE):

$$d\mathbf{x} = -\frac{\beta_t}{2}\mathbf{x}dt + \sqrt{\beta_t}d\mathbf{w}, \quad t \in [0, T] \quad (3)$$

where $\beta_t > 0$ (monotonically increasing) controls noise addition, and w is a Wiener process. At $t = T$, $\mathbf{x}_T \sim \mathcal{N}(0, I)$. In the reverse process, samples are generated by solving the reverse-time SDE:

$$d\mathbf{x} = \left(-\frac{\beta_t}{2}\mathbf{x} - \beta_t \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right) dt + \sqrt{\beta_t}d\bar{\mathbf{w}}, \quad (4)$$

where $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ (score function) is approximated by a neural network $\mathbf{s}_\theta(\mathbf{x}, t)$.

Zero-shot diffusion methods enable training/data-free inverse problem solving. For HSI-MSI fusion, we use posterior sampling to reconstruct HR-HSI $\mathcal{Z}_0$ from observations $\mathcal{X}$ (HR-MSI) and $\mathcal{Y}$ (LR-HSI). The posterior distribution is:

$$p(\mathcal{Z}_0|\mathcal{X}, \mathcal{Y}) \propto p(\mathcal{X}, \mathcal{Y}|\mathcal{Z}_0) \cdot p(\mathcal{Z}_0). \quad (5)$$

Then, the reverse process of the diffusion sampler can be expressed as

$$d\mathcal{Z} = \left[-\frac{\beta_t}{2}\mathcal{Z} - \beta_t(\nabla_{\mathcal{Z}_t} \log p_t(\mathcal{Z}_t) + \nabla_{\mathcal{Z}_t} \log p_t(\mathcal{X}, \mathcal{Y}|\mathcal{Z}_t)) \right] dt + \sqrt{\beta_t}d\bar{\mathbf{w}}. \quad (6)$$

According to **Theorem 1** in [25], the gradient of the likelihood can be computed as

$$\nabla_{\mathcal{Z}_t} \log p_t(\mathcal{X}, \mathcal{Y}|\mathcal{Z}_t) \simeq \nabla_{\mathcal{Z}_t} \log p(\mathcal{X}, \mathcal{Y}|\hat{\mathcal{Z}}_0), \quad (7)$$

where $\hat{\mathcal{Z}}_0$ takes the global maximum of $p(\mathcal{Z}_0|\mathcal{Z}_t)$ and it can be computed by

$$\hat{\mathcal{Z}}_0 := \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathcal{Z}_t + (1 - \bar{\alpha}_t)\mathbf{s}_\theta(\mathcal{Z}_t, t)). \quad (8)$$

Given that the observations $\mathcal{X}$ and $\mathcal{Y}$ are independent given $\hat{\mathcal{Z}}_0$. Thus, the likelihood function can be expressed as

$$\begin{aligned} p(\mathcal{X}, \mathcal{Y}|\hat{\mathcal{Z}}_0) &= p(\mathcal{X}|\hat{\mathcal{Z}}_0)p(\mathcal{Y}|\hat{\mathcal{Z}}_0) \\ &= \frac{1}{\sqrt{(2\pi)^n \sigma_1^{2n}}} \exp \left[-\frac{\|\mathcal{X} - \hat{\mathcal{Z}}_0 \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\|_2^2}{2\sigma_1^2} \right] * \\ &\quad \frac{1}{\sqrt{(2\pi)^n \sigma_2^{2n}}} \exp \left[-\frac{\|\mathcal{Y} - \hat{\mathcal{Z}}_0 \times_3 \mathbf{S}\|_2^2}{2\sigma_2^2} \right]. \end{aligned} \quad (9)$$

With **Theorem 1** [25], we have the posterior gradient

$$\begin{aligned} \nabla_{\mathcal{Z}_t} \log p_t(\mathcal{Z}_t|\mathcal{X}, \mathcal{Y}) &\simeq \mathbf{s}_{\theta^*}(\mathcal{Z}_t, t) \\ &\quad - \rho \nabla_{\mathcal{Z}_t} (\|\mathcal{X} - \hat{\mathcal{Z}}_0 \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\|_2^2 + \lambda \|\mathcal{Y} - \hat{\mathcal{Z}}_0 \times_3 \mathbf{S}\|_2^2). \end{aligned} \quad (10)$$

Algorithm 1 Diffusion Posterior Sampling for HSI and MSI Fusion

Input: LR-HSI $\mathcal{X}$, HR-MSI $\mathcal{Y}$, $\{\tilde{\sigma}_t\}_{t=1}^N$.

1: $\mathcal{Z}_N \sim \mathcal{N}(0, I)$

2: **for** $t = N$ to 1 **do**

3: $\hat{\mathbf{s}} \leftarrow \mathbf{s}_\theta(\mathcal{Z}_t, t) = -\frac{1}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\mathcal{Z}_t, t)$

4: $\hat{\mathcal{Z}}_0 \leftarrow \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathcal{Z}_t + (1 - \bar{\alpha}_t)\hat{\mathbf{s}})$

5: $\epsilon \sim \mathcal{N}(0, I)$

6: $\mathcal{Z}'_{t-1} \leftarrow \frac{\sqrt{\bar{\alpha}_t(1-\bar{\alpha}_{t-1})}}{1-\bar{\alpha}_t}\mathcal{Z}_t + \frac{\sqrt{\bar{\alpha}_{t-1}\beta_t}}{1-\bar{\alpha}_t}\hat{\mathcal{Z}}_0 + \tilde{\sigma}_t\epsilon$

7: $\mathcal{Z}_{t-1} \leftarrow \mathcal{Z}'_{t-1} - \rho \nabla_{\mathcal{Z}_t} (\|\mathcal{X} - \hat{\mathcal{Z}}_0 \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\|_2^2 + \lambda \|\mathcal{Y} - \hat{\mathcal{Z}}_0 \times_3 \mathbf{S}\|_2^2)$

8: **end for**

Output: Reconstructed HR-HSI $\mathcal{Z}_0$

This enables efficient sampling of HR-HSI that respects both physical constraints $(\mathcal{X}, \mathcal{Y})$ and data priors. The DPS algorithm for HSI and MSI fusion is summarized in **Algorithm 1**.

C. Coupled Diffusion Posterior Sampling

In the diffusion posterior sampling, the score function should be trained. However, the score function $\mathbf{s}_\theta(\mathcal{Z}_t, t)$ is trained using the HR-HSI which can be leveraged to estimate the distribution of the HR-HSI. In general, the HR-HSI is the high-dimensional data whose distribution is hard to estimate and the training requires a large amount of real HR-HSIs. However, in real application, seldom HR-HSIs are available for training. Actually, we only have the LR-HSI and HR-MSI in the HSI and MSI fusion. Thus, the score function of the HR-HSI is not easy to get.

To solve this problem, many methods [46] have been proposed. A common approach involves mapping high-dimensional data to a lower-dimensional space and then applying the diffusion model within this reduced space. In this paper, we propose a coupled diffusion model which decomposes the high-dimensional HR-HSI data into two low-dimensional components. Therefore, we can replace the score function in the original high-dimensional space with two score

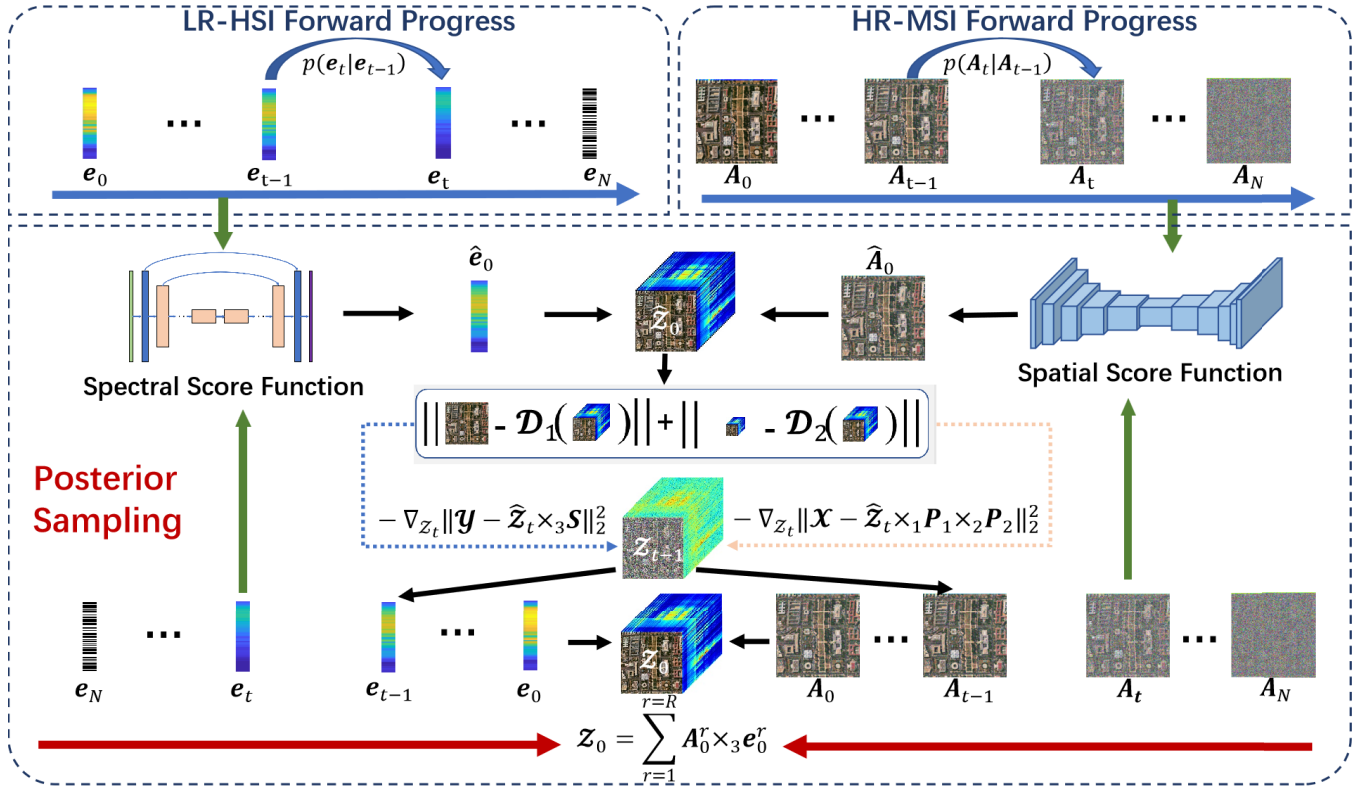


Fig. 1. Overall framework of our proposed CDPS method. The forward processes of LR-HSI and HR-MSI are respectively used to derive the spectral score function and the spatial score function. During the sampling procedure, the spectral component $\mathbf{E}_0$ and spatial component $\mathbf{A}_0$ are progressively reconstructed from pure noise. The final HR-HSI $\mathbf{Z}_0$ is reconstructed by combining the estimated $\mathbf{E}_0$ and $\mathbf{A}_0$ components using the linear mixing model $\mathbf{Z}_0 = \mathbf{A}_0 \times_3 \mathbf{E}_0$. $\mathcal{D}_1$ and $\mathcal{D}_2$ denote the spectral and spatial downsampling operators, respectively.

functions in the low-dimensional spaces. The flowchart of the proposed method is presented in Fig. 1.

In fact, each pixel in the HR-HSI can be represented as a sparse linear combination of spectral atoms from a learned dictionary—a paradigm known as the dictionary learning model. Mathematically, we express this as:

$$\mathbf{Z} = \mathbf{A} \times_3 \mathbf{E}. \quad (11)$$

Here, $\mathbf{E} \in \mathbb{R}^{R \times B}$ is the dictionary matrix whose rows $\mathbf{e}^r$ are spectral atoms representing fundamental materials. $\mathbf{A} \in \mathbb{R}^{W \times H \times R}$ contains the spatial coefficient maps weighting each atom. The model equivalently decomposes as:

$$\mathbf{Z} = \sum_{r=1}^R \mathbf{A}^r \times_3 \mathbf{e}^r, \quad (12)$$

where $\mathbf{A}^r \in \mathbb{R}^{W \times H \times 1}$ is the r -th coefficient map or the abundance map and $\mathbf{e}^r \in \mathbb{R}^{1 \times B}$ is its corresponding dictionary atom. In conventional dictionary learning, dictionaries are often overcomplete ($R \gg B$), requiring sparsity constraints on coefficients to resolve redundancy. However, in hyperspectral unmixing under the Linear Mixing Model (LMM), the dictionary has a physical interpretation: its atoms $\mathbf{E} = [\mathbf{e}^1, \dots, \mathbf{e}^R]^T$ represent pure spectral signatures of distinct materials (endmembers). Crucially, the number of endmembers R is typically smaller than the spectral dimension B ($R \ll B$), resulting in a compact (non-redundant) dictionary.

From the above Equation, we can see that the HR-HSI can be decomposed into two components—abundance map and the corresponding endmember where the abundance map is a single band image and the endmember is a spectrum. Both the abundance map and endmember are in a lower-dimensional space, which makes it easy to train the diffusion model. Therefore, the task shifts to training the score function for the abundance map and endmembers. In [26], two untrained Unets are leveraged to generate the abundance map and endmembers. However, the untrained net cannot characterize the distribution of the abundance and endmembers, which may lead to limited improvement.

In this paper, we propose an unsupervised method to train the two score functions. In the observed HR-MSI, the spatial resolution matches that of the target HR-HSI. Critically, this spatial equivalence means the abundance maps $\mathbf{A}^r$ which define the spatial distribution of materials—manifest identically in both modalities. Therefore, the spatial distribution within any HR-MSI band inherently reflects the spatial structure of these underlying abundance maps. We leverage this relationship to train the spatial score function $s_a(\mathbf{A}_t^r, t)$ of the abundance map with $\mathcal{Y}$. For the spectral component, we train the spectral score function $s_b(\mathbf{e}_t^r, t)$ using spectral vectors from the LR-HSI $\mathcal{X}$ to learn the spectral characteristics of the scene's materials. This learned prior guides the generation of endmembers $\mathbf{E}$ to ensure they constitute physically meaningful spectral signatures that can explain the observed data through

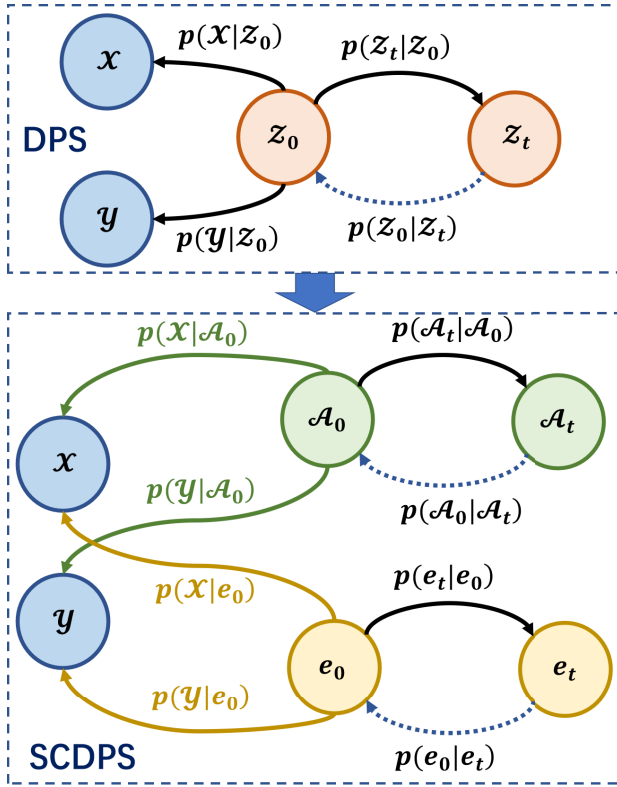


Fig. 2. Graphic Model of the DPS and the proposed CDPS. (Top) DPS estimates a single latent variable $\mathbf{Z}_0$ (the target data) conditioned on the observations $\mathbf{X}$ and $\mathbf{Y}$. (Bottom) CDPS decomposes the problem, modeling two independent latent variables—the spectral component $\mathbf{E}_0$ and the spatial component $\mathbf{A}_0$ which are conditioned on two observations, LR-HSI and HR-MSI..

the linear mixing model. The two trained score functions are then physically coupled through the linear mixing model to generate the HR-HSI:

$$\mathbf{Z}_0 = \sum_{r=1}^R \mathbf{A}_0^r \times_3 \mathbf{e}_0^r, \quad (13)$$

where $\mathbf{A}_0^r$ and $\mathbf{e}_0^r$ are generated by the spatial diffusion network and spectral diffusion network.

Since $\mathbf{Z}_0 = \mathbf{A}_0 \times_3 \mathbf{E}_0$, $\mathbf{A}_0$ and $\mathbf{E}_0$ are independent, the posterior distribution of the HR-HSI can be rewritten as

$$p(\mathbf{A}_0, \mathbf{E}_0 | \mathbf{X}, \mathbf{Y}) \propto p(\mathbf{X}, \mathbf{Y} | \mathbf{Z}_0) p(\mathbf{A}_0) p(\mathbf{E}_0). \quad (14)$$

And we can construct two separate reverse diffusion processes with a similar form:

$$d\mathbf{A}^r = \left[-\frac{\beta_t}{2} \mathbf{A}^r - \beta_t (\nabla_{\mathbf{A}^r} \log p_t(\mathbf{A}^r) + \nabla_{\mathbf{A}^r} \log p_t(\mathbf{X}, \mathbf{Y} | \mathbf{A}^r, \mathbf{e}_t^r)) \right] dt + \sqrt{\beta_t} d\mathbf{w}, \quad (15)$$

$$d\mathbf{e}^r = \left[-\frac{\beta_t}{2} \mathbf{e}^r - \beta_t (\nabla_{\mathbf{e}^r} \log p_t(\mathbf{e}^r) + \nabla_{\mathbf{e}^r} \log p_t(\mathbf{X}, \mathbf{Y} | \mathbf{A}^r, \mathbf{e}_t^r)) \right] dt + \sqrt{\beta_t} d\mathbf{w}. \quad (16)$$

Then, we can construct the spectral and spatial reverse diffusion process as:

$$\nabla_{\mathbf{A}^r} \log p_t(\mathbf{X}, \mathbf{Y} | \mathbf{A}^r, \mathbf{e}_t^r) \simeq \nabla_{\mathbf{A}^r} \log p(\mathbf{X}, \mathbf{Y} | \hat{\mathbf{A}}_0^r, \hat{\mathbf{e}}_0^r), \quad (17)$$

$$\nabla_{\mathbf{e}^r} \log p_t(\mathbf{X}, \mathbf{Y} | \mathbf{A}^r, \mathbf{e}_t^r) \simeq \nabla_{\mathbf{e}^r} \log p(\mathbf{X}, \mathbf{Y} | \hat{\mathbf{A}}_0^r, \hat{\mathbf{e}}_0^r), \quad (18)$$

where $\hat{\mathbf{A}}_0^r, \hat{\mathbf{e}}_0^r$ takes the global maximum of $p(\mathbf{A}_0^r | \mathbf{A}^r)$, and $p(\mathbf{e}_0 | \mathbf{e}_t)$ similar to Eq. (7) respectively. The graphic model of the proposed method is presented in Fig. 2.

It is essential to emphasize that the endmembers $\mathbf{E}_0$ are not direct samples from the distribution $p(\mathbf{X})$ learned from the LR-HSI. Rather, the learned prior $p(\mathbf{X})$ captures the spectral characteristics of the scene, which then constrains $\mathbf{E}_0$ to represent fundamental spectral components that can linearly combine to reconstruct the observed spectral mixtures. This ensures physical consistency while respecting the dimensional differences between the pure endmembers $\mathbf{E}_0$ and the mixed observations $\mathbf{X}$.

Finally, the posterior gradient for the spatial and spectral mode are as follows.

$$\begin{aligned} \nabla_{\mathbf{A}^r} \log p_t(\mathbf{A}^r | \mathbf{X}, \mathbf{Y}) &\simeq \mathbf{s}_{\alpha^*}(\mathbf{A}^r, t) \\ &- \rho_1 \nabla_{\mathbf{A}^r} (\|\mathbf{X} - (\sum_{r=1}^{r=R} \hat{\mathbf{A}}_0^r \times_3 \hat{\mathbf{e}}_0^r) \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\|_2^2 \\ &+ \lambda_1 \|\mathbf{Y} - (\sum_{r=1}^{r=R} \hat{\mathbf{A}}_0^r \times_3 \hat{\mathbf{e}}_0^r) \times_3 \mathbf{S}\|_2^2), \end{aligned} \quad (19)$$

$$\begin{aligned} \nabla_{\mathbf{e}^r} \log p_t(\mathbf{e}^r | \mathbf{X}, \mathbf{Y}) &\simeq \mathbf{s}_{\beta^*}(\mathbf{e}^r, t) \\ &- \rho_2 \nabla_{\mathbf{e}^r} (\|\mathbf{X} - (\sum_{r=1}^{r=R} \hat{\mathbf{A}}_0^r \times_3 \hat{\mathbf{e}}_0^r) \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\|_2^2 \\ &+ \lambda_2 \|\mathbf{Y} - (\sum_{r=1}^{r=R} \hat{\mathbf{A}}_0^r \times_3 \hat{\mathbf{e}}_0^r) \times_3 \mathbf{S}\|_2^2). \end{aligned} \quad (20)$$

The reconstructed HR-HSI can be written as $\mathbf{Z}_0 = \mathbf{A}_0 \times \mathbf{E}_0 = \sum_{r=1}^{r=R} \mathbf{A}_0^r \times_3 \mathbf{e}_0^r$.

Algorithm 2 Coupled Diffusion Posterior Sampling for Unsupervised HSI and MSI Fusion

Input: N , LR-HSI $\mathbf{X}$, HR-MSI $\mathbf{Y}$, $\{\tilde{\sigma}_t\}_{t=1}^N$, endmembers number R , $\rho_1, \lambda_1, \rho_2, \lambda_2, \rho_3, \lambda_3$.

- 1: $\mathbf{A}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\mathbf{E}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: Training the spectral score function $\mathbf{s}_{\alpha^*}(\mathbf{A}_t^r, t)$ with $\mathbf{X}$, training the spatial score function $\mathbf{s}_{\beta^*}(\mathbf{e}_t^r, t)$ with $\mathbf{Y}$
- 3: **for** $t = N$ to 1 **do**
- 4: **for** $r = 1$ to R **do**
- 5: $\hat{\mathbf{s}}_\alpha \leftarrow \mathbf{s}_{\alpha^*}(\mathbf{A}_t^r, t)$
- 6: $\hat{\mathbf{s}}_\beta \leftarrow \mathbf{s}_{\beta^*}(\mathbf{e}_t^r, t)$
- 7: $\hat{\mathbf{A}}_0^r \leftarrow \frac{1}{\sqrt{\alpha_t}} (\mathbf{A}_t^r + (1 - \alpha_t) \hat{\mathbf{s}}_\alpha)$
- 8: $\hat{\mathbf{e}}_0^r \leftarrow \frac{1}{\sqrt{\beta_t}} (\mathbf{e}_t^r + (1 - \beta_t) \hat{\mathbf{s}}_\beta)$
- 9: $\epsilon_\alpha \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\epsilon_\beta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 10: $\mathbf{A}_{t-1}^{r'} \leftarrow \frac{\sqrt{\alpha_t(1-\alpha_{t-1})}}{1-\alpha_t} \mathbf{A}_t^r + \frac{\sqrt{\alpha_{t-1}\beta_t}}{1-\alpha_t} \hat{\mathbf{A}}_0^r + \tilde{\sigma}_t \epsilon_\alpha$
- 11: $\mathbf{e}_{t-1}^{r'} \leftarrow \frac{\sqrt{\beta_t(1-\beta_{t-1})}}{1-\beta_t} \mathbf{e}_t^r + \frac{\sqrt{\beta_{t-1}\alpha_t}}{1-\beta_t} \hat{\mathbf{e}}_0^r + \tilde{\sigma}_t \epsilon_\beta$
- 12: $\mathbf{A}_{t-1}^r \leftarrow \mathbf{A}_{t-1}^{r'} - \rho_1 \nabla_{\mathbf{A}^r} (\|\mathbf{X} - (\sum_{r=1}^{r=R} \hat{\mathbf{A}}_0^r \times_3 \hat{\mathbf{e}}_0^r) \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\|_2^2 + \lambda_1 \|\mathbf{Y} - (\sum_{r=1}^{r=R} \hat{\mathbf{A}}_0^r \times_3 \hat{\mathbf{e}}_0^r) \times_3 \mathbf{S}\|_2^2)$
- 13: $\mathbf{e}_{t-1}^r \leftarrow \mathbf{e}_{t-1}^{r'} - \rho_2 \nabla_{\mathbf{e}^r} (\|\mathbf{X} - (\sum_{r=1}^{r=R} \hat{\mathbf{A}}_0^r \times_3 \hat{\mathbf{e}}_0^r) \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\|_2^2 + \lambda_2 \|\mathbf{Y} - (\sum_{r=1}^{r=R} \hat{\mathbf{A}}_0^r \times_3 \hat{\mathbf{e}}_0^r) \times_3 \mathbf{S}\|_2^2)$
- 14: **end for**
- 15: **end for**

Output: HR-HSI $\mathbf{Z}_0 = \mathbf{A}_0 \times \mathbf{E}_0 = \sum_{r=1}^{r=R} \mathbf{A}_0^r \times_3 \mathbf{e}_0^r$

Finally, the CDPS algorithm for HSI and MSI fusion is summarized in Algorithm 2.

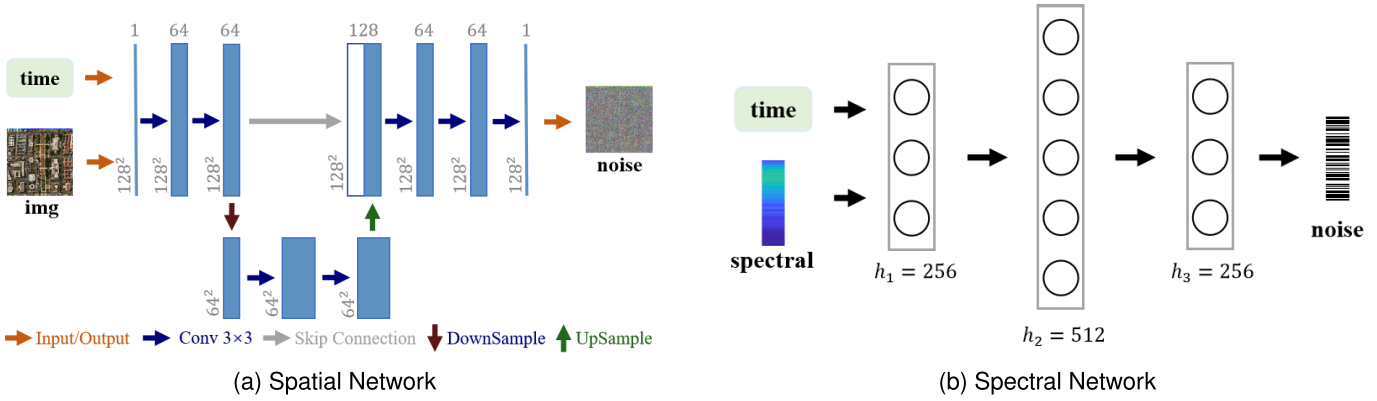


Fig. 3. The detailed structures of spatial and spectral networks.

D. Training of The Lightweight Spatial and Spectral Priors

In the traditional diffusion networks, the signal to be generated always has a high dimensionality. For example, in [47], the generated image has a resolution of 1024×1024 . To estimate the probability distribution of such a high-dimensional signal is very difficult. To solve this problem, deep and complex neural networks are designed in noise estimation. Thus, the training of the score function requires a long time. To reduce the training time, researchers usually map the high-dimensional signal to a lower-dimensional space via variational autoencoder (VAE). However, the mapping based on VAE is lossy and unexplainable.

HSI has higher spectral resolution than natural images, thus the dimension of HSI is even higher. Directly training the HSI is not feasible and is time-consuming. In this paper, we propose to decompose the HSI into two lower-dimensional spaces by a physically driven unmixing model. Thus, instead of learning the distribution of the original high-dimensional HSI, we only need to learn two simple and low-dimensional signals - single band spatial signal and single-pixel spectral signal. Due to the spatial and spectral signals being low-dimensional, we can use simple networks to learn the score functions. Fig. 3 shows the two networks. In the spatial network, we adopt a lightweight U-Net architecture. It consists of a single downsampling layer, an intermediate layer, and a corresponding upsampling layer, where each layer is composed of two basic convolutional blocks. In the spectral network, we employ a fully connected network (FCN) composed of three successive linear layers.

Since the inputs are reduced to low-dimensional signals, the training does not require large external datasets. Instead, we directly use the available observations to construct the training data. Specifically, taking the Pavia dataset as an example, during the training of the spatial network, single-band image patches are cropped from the observed HR-MSI to form input tensors of shape $(1, 128, 128)$, which are fed into the spatial network shown in Fig. 3(a). For the spectral network, individual pixel spectra are extracted from the observed LR-HSI, resulting in input tensors of shape $(1, 103)$ for the spectral network shown in Fig. 3(b). These observed signals therefore serve simultaneously as both the training data and the inference inputs.

The spectral and spatial priors are learned directly from the given observations of the test scene itself, without relying on any external paired HR-HSI datasets. This training-data-free paradigm is the key to the practical applicability of our framework in scenarios where no ground-truth HR-HSI is available for training.

IV. EXPERIMENTAL ANALYSIS

A. Datasets and Implementation Details

In our experiments, we utilized several publicly available datasets to validate the performance of our model, namely CAVE, Pavia University, and Chikusei. The CAVE dataset contains 32 indoor HSIs captured by a cooled CCD camera. Each image measures 512×512 pixels across 31 spectral bands, spanning wavelengths from 400 to 700 nm with a 10 nm step size. We applied bicubic interpolation to downsample the images to 256×256 pixels, maintaining the 31 spectral bands, to serve as ground truth.

The Pavia University dataset is a subset of hyperspectral data collected over Pavia, Italy, in 2003 by the Reflective Optics Spectrographic Imaging System (ROSIS-03), an airborne hyperspectral imaging instrument. Capturing data across 115 bands within the wavelength range of $0.43 - 0.86 \mu\text{m}$, with a spatial resolution of 1.3 m, the instrument provided valuable insights. Due to noise, 12 bands were discarded, resulting in 103 spectral bands typically used. We extracted a $256 \times 256 \times 103$ patch from the original image, sized $610 \times 340 \times 103$, to serve as ground truth.

The Chikusei dataset consists of aerial hyperspectral imagery acquired by the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Ibaraki, Japan. The dataset contains 128 spectral bands covering the range from 363 nm to 1018 nm, with spatial dimensions of 2517×2335 pixels. From this dataset, we removed noisy bands, resulting in 110 bands. Then, we extracted an image of size 256×256 pixels across these 110 spectral bands to serve as the ground truth.

To generate paired HR-MSI and LR-HSI data, we synthesized the HR-MSI by applying a simulated spectral response function (SRF) to the ground truth HSI. This yielded HR-MSI with 3 bands for CAVE, 4 for Pavia, and 5 for Chikusei.

LR-HSI was obtained by applying bicubic interpolation to the ground truth HSI, with a downsampling factor of 4.

To train the spatial model, we extracted grayscale maps from each channel of HR-MSI, producing 128×128 images with a 64-pixel overlap. This process yielded 27, 36, and 45 training images for the three datasets respectively. The batch sizes were set to 24, 32, and 32, with a total of 20k iterations. When training the spectral model, we extracted the spectral curves corresponding to each pixel in the LR-HSI as training samples. Given the varied spectral complexity across the datasets, we adjusted the number of iterations to 300k, 500k, and 600k respectively. The batch size remained constant at 128. The main experiments are conducted using the original datasets, leveraging their inherent sensor-induced noise without any controlled introduction of additional artifacts. Supplementary experiments involving the deliberate injection of synthetic noise to evaluate robustness are reported in Section H.

We conducted these experiments using the PyTorch framework on an NVIDIA GeForce RTX 4090 GPU, utilizing the Adam optimizer with a maximum learning rate of 0.0001. Additionally, we set the total number of time steps to 1000 and incorporated an exponential noise schedule for β [48].

B. Evaluation Metrics

For conducting quantitative evaluation, we employed five commonly used metrics: peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM) [49], error relative global adimensionnelle de synthese (ERGAS) [50], structural similarity (SSIM) [51], and universal image quality index (UIQI) [52].

1) *PSNR*: PSNR is a metric for assessing the quality of image reconstruction, which quantifies the ratio of signal to noise between the original image and the reconstructed image. The PSNR of the k th band is defined as

$$PSNR = 10 \log_{10} \left(\frac{\max(\mathbf{R}_k)^2}{\frac{1}{HW} \|\mathbf{R}_k - \mathbf{Z}_k\|_2^2} \right), \quad (21)$$

where $\mathbf{R}_k$ and $\mathbf{Z}_k$ denote the target HR-HSI and reconstructed HR-HSI at k th band, respectively. The final PSNR value is obtained by averaging across all channels.

2) *SAM*: SAM is used to evaluate the classification quality of remote sensing images. It measures the similarity between two pixels in feature space by calculating the angle between them. A smaller angle indicates a higher similarity in spectral characteristics between the two pixels. It is defined as

$$SAM = \arccos \left(\frac{\langle \mathbf{R}(i, j), \mathbf{Z}(i, j) \rangle}{\|\mathbf{R}(i, j)\|_2 \|\mathbf{Z}(i, j)\|_2} \right), \quad (22)$$

where $\mathbf{R}(i, j)$ and $\mathbf{Z}(i, j)$ represent the spectral vectors of the target HR-HSI and the reconstructed HR-HSI at position (i, j) . $\langle \mathbf{R}(i, j), \mathbf{Z}(i, j) \rangle$ represents the dot product between vector $\text{textbf{R}}(i, j)$ and vector $\mathbf{Z}(i, j)$.

3) *ERGAS*: ERGAS combines the ratio of root mean square error and the mean value of the image to provide a comprehensive global image quality measure, which is defined as

$$ERGAS = \frac{100}{r} \sqrt{\frac{1}{B} \sum_{k=1}^B \frac{\|\mathbf{R}_k - \mathbf{Z}_k\|_2^2}{\mu^2(\mathbf{R}_k)}}, \quad (23)$$

TABLE I

TEST PERFORMANCE ON THE CAVE DATASET. THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE WITH UNDERLINE

Methods	PSNR $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	SSIM $\uparrow$	UIQI $\uparrow$
CNMF	38.88	8.44	2.71	0.969	0.946
HySure	39.96	6.70	<u>2.33</u>	0.977	0.977
CuCaNet	37.13	6.52	<u>3.77</u>	0.971	0.973
HyCoNet	37.11	8.84	3.67	0.971	0.964
MIAE	39.98	<u>5.12</u>	2.51	<u>0.982</u>	0.983
PLRDiff	35.83	16.47	4.30	0.952	0.935
S ² CycleDiff	38.06	7.68	3.36	0.975	0.950
CDPS	40.88	5.02	2.18	0.986	0.986

where r represents the downsampling factor, and $\mu(\mathbf{R}_k)$ refers to the mean value of the reconstructed HR-HSI at the k th spectral band. A lower ERGAS value indicates a better fusion effect.

4) *SSIM*: SSIM considers luminance similarity, contrast similarity, and structural similarity by calculating the weighted product of these similarities to obtain the final similarity index. It is defined as

$$SSIM = \frac{(2\mu(\mathbf{R}_k)\mu(\mathbf{Z}_k) + c_1)(2\sigma(\mathbf{R}_k, \mathbf{Z}_k) + c_2)}{(\mu^2(\mathbf{R}_k) + \mu^2(\mathbf{Z}_k) + c_1)(\sigma^2(\mathbf{R}_k) + \sigma^2(\mathbf{Z}_k) + c_2)}, \quad (24)$$

where $\sigma^2(\mathbf{R}_k)$ and $\sigma^2(\mathbf{Z}_k)$ denote the variances of the target HR-HSI and the reconstructed HR-HSI at k th spectral band. $\sigma(\mathbf{R}_k, \mathbf{Z}_k)$ is the covariance between the target HR-HSI and the reconstructed HR-HSI at k th spectral band, c_1 and c_2 are constants. The value of SSIM varies between -1 and 1, where 1 indicates that two images are identical.

5) *UIQI*: UIQI integrates luminance, contrast, and structural information by measuring the similarity between two images through the comparison of their statistical differences. It can be defined as

$$UIQI = \frac{4\mu(\mathbf{R}_k)\mu(\mathbf{Z}_k)\sigma(\mathbf{R}_k, \mathbf{Z}_k)}{(\mu^2(\mathbf{R}_k) + \mu^2(\mathbf{Z}_k))(\sigma^2(\mathbf{R}_k) + \sigma^2(\mathbf{Z}_k))}. \quad (25)$$

The value of UIQI typically ranges from 0 to 1, where a value closer to 1 indicates higher similarity and better quality between the two images.

C. Comparison With State-of-the-Art Methods

To illustrate the effectiveness of our proposed method, we have chosen two conventional methods and five deep learning approaches for a comprehensive comparison. The conventional methods are CNMF [32] and HySure [34]. The DL-based methodologies included neural network-based subspace approaches, such as CuCaNet [53], HyCoNet [54], and MIAE [55], alongside diffusion-based methods like PLRDiff [48] and S²CycleDiff [45]. For fairness, we uniformly set the number of endmembers R to 8 for all subspace-related methods, as detailed in IV-F1.

1) *Results on CAVE*: Firstly, we conducted experiments on the CAVE dataset using our proposed method and the aforementioned comparative methods. The results, summarized in Table I, indicate that our method outperforms all others across

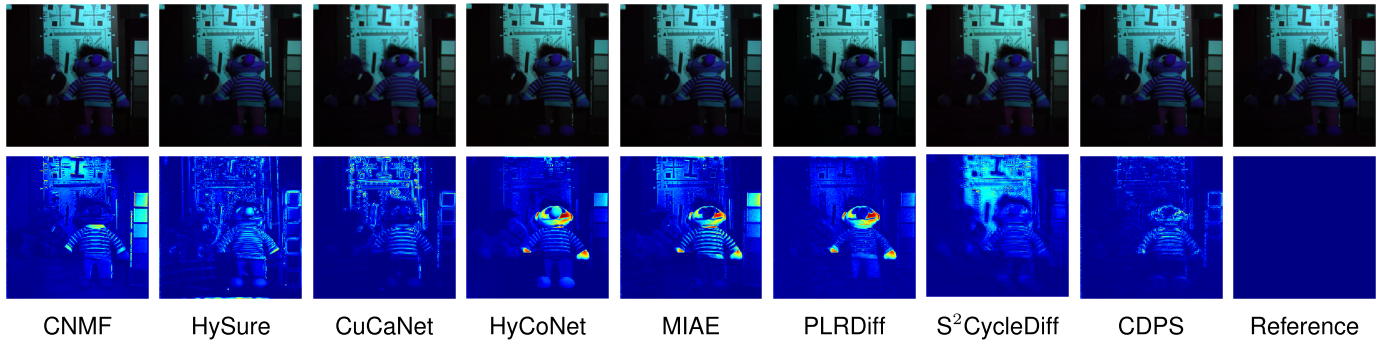


Fig. 4. Visual results and reconstruction error maps obtained by different methods on the CAVE dataset. The darker error maps indicate better performance.

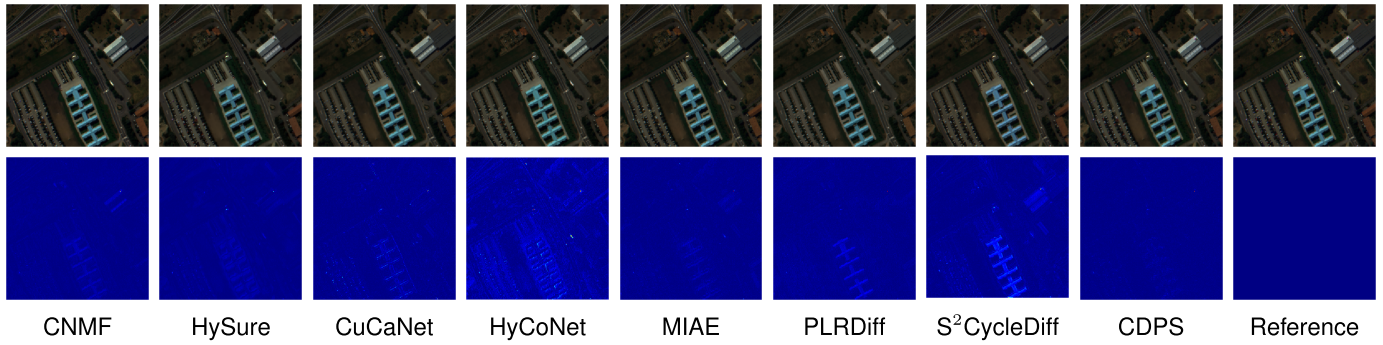


Fig. 5. Visual results and reconstruction error maps obtained by different methods on the Pavia dataset. The darker error maps indicate better performance.

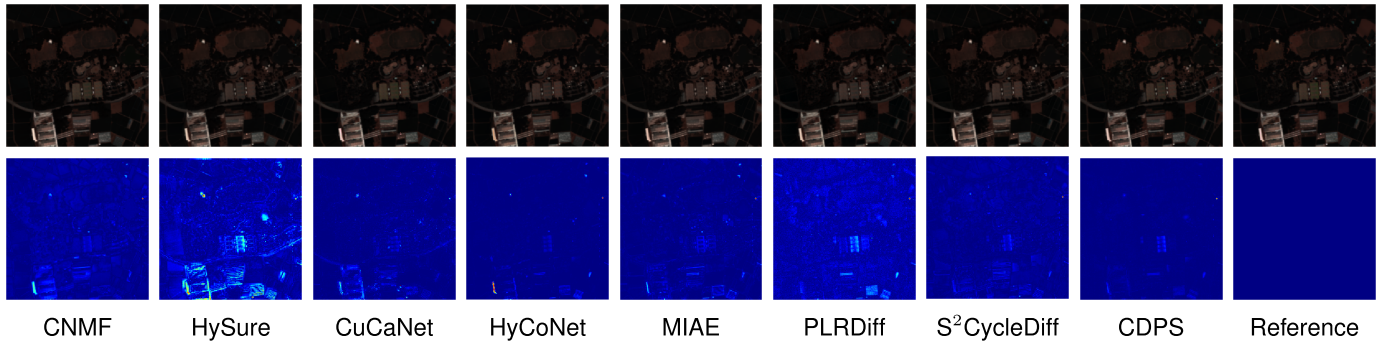


Fig. 6. Visual results and reconstruction error maps obtained by different methods on Chikusei dataset. The darker error maps indicate better performance.

every metric. Notably, it achieves a PSNR value nearly 1 dB higher than the second-ranked MIAE method, highlighting the exceptional fusion performance of our approach on the CAVE dataset. In comparison, the diffusion-based method PLRDiff exhibits inferior performance, likely due to the pre-trained model not being trained on the CAVE dataset. Similarly, S²CycleDiff, as a supervised learning approach based on diffusion models, still lags significantly behind our method in performance. For a visual representation of the fusion effect, Fig. 4 displays RGB images of the reconstructed HR-HSI from each method, accompanied by corresponding error maps. These visuals confirm that our method achieves the best fusion performance. Additionally, as shown in Fig. 7, the HR-HSI reconstructed by our method exhibits the highest spectral fidelity, with minimal spectral distortion, further demonstrating the superior effectiveness of our method in image fusion on the CAVE dataset.

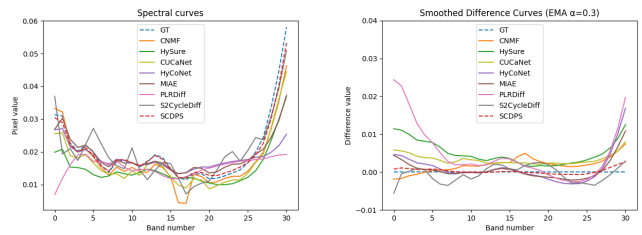


Fig. 7. Example spectral curves and difference curves on a selected pixel value of CAVE datasets. The difference curve was smoothed using an exponential weighted moving average for better interpretability.

2) *Results on Pavia:* Secondly, Table II presents the experimental results of various methods on the Pavia dataset, while Fig. 5 provides the fused RGB images along with their corresponding error maps. It is evident from Table II that

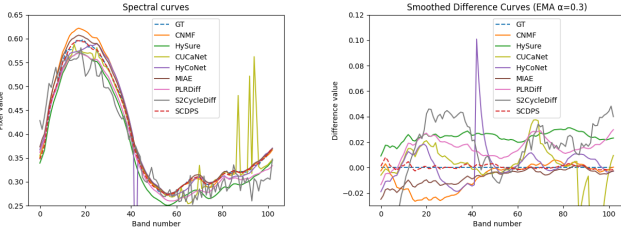


Fig. 8. Example spectral curves and difference curves on a selected pixel value of Pavia datasets. The difference curve was smoothed using an exponential weighted moving average for better interpretability.

TABLE II

TEST PERFORMANCE ON THE PAVIA DATASET. THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE WITH UNDERLINE

Methods	PSNR↑	SAM↓	ERGAS↓	SSIM↑	UIQI↑
CNMF	39.89	2.80	1.77	0.975	0.995
HySure	38.01	3.12	2.27	0.972	0.994
CuCaNet	38.28	9.21	3.78	0.949	0.980
HyCoNet	39.19	5.28	3.50	0.964	0.986
MIAE	<u>41.64</u>	<u>2.63</u>	<u>1.63</u>	<u>0.976</u>	0.997
PLRDiff	40.04	3.25	1.98	0.974	0.995
S ² CycleDiff	38.04	3.12	2.22	0.970	0.995
CDPS	41.95	2.58	1.60	0.979	0.997

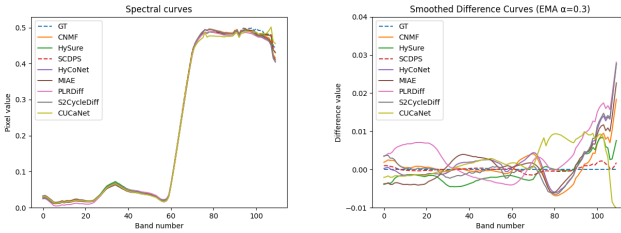


Fig. 9. Example spectral curves and difference curves on a selected pixel value of Chikusei datasets. The difference curve was smoothed using an exponential weighted moving average for better interpretability.

our proposed method achieves the best performance among all methods for all five evaluation metrics, demonstrating excellent spatial, spectral, and structural image reconstruction. The MIAE method ranks second overall, while diffusion-based methods such as PLRDiff and S²CycleDiff lag behind significantly, especially in spectral and spatial accuracy, as evidenced by their higher SAM and ERGAS scores. From Fig. 5, the fusion effects of various methods on the Pavia dataset are satisfactory, but CDPS achieves the smallest deviation as observed from the error maps. Meanwhile, Fig. 8 confirms the outstanding performance of our proposed method, as the reconstructed images obtained by CDPS exhibit the smallest spectral differences.

3) *Results on Chikusei*: Thirdly, we conducted comparative experiments on the Chikusei dataset to further demonstrate the universality and effectiveness of the proposed method. The experimental results, summarized in Table III, show that CDPS consistently outperforms all other methods across key metrics, showcasing exceptional image reconstruction quality with minimal spectral and spatial distortions. While HyCoNet

TABLE III

TEST PERFORMANCE ON THE CHIKUSEI DATASET. THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE WITH UNDERLINE

Methods	PSNR↑	SAM↓	ERGAS↓	SSIM↑	UIQI↑
CNMF	41.19	2.35	2.15	0.984	0.992
HySure	41.69	1.97	2.18	0.985	0.994
CuCaNet	40.06	1.74	2.63	0.974	0.990
HyCoNet	<u>43.84</u>	<u>1.71</u>	<u>2.05</u>	0.987	0.995
MIAE	43.71	1.72	2.19	<u>0.988</u>	0.996
PLRDiff	40.94	2.07	2.30	0.981	0.991
S ² CycleDiff	40.83	2.15	2.44	0.979	0.990
CDPS	44.26	1.59	1.87	0.990	0.996

and MIAE deliver competitive results, particularly in PSNR, SSIM, and SAM, they still fall short of CDPS's overall performance. On the other hand, diffusion-based methods like PLRDiff perform weaker across most metrics, particularly in spectral and structural integrity. As shown in Fig. 6, most methods' error maps reveal significant deviations, except for HyCoNet and CDPS, with CDPS exhibiting smaller errors than HyCoNet. Furthermore, from the perspective of spectral curves in Fig. 9, CDPS exhibits the smallest curve deviations, further underscoring the superior fusion performance of the proposed method.

D. Experimental Results on Real-World Data

We have already conducted experiments on three simulated datasets, which demonstrate the superior fusion performance of our model. To further validate its practical applicability, we perform additional experiments on real data. Specifically, we adopt the DFC2018 Houston dataset to evaluate the performance of our proposed method in real-world scenarios. Fig. 10 presents the observed LR-HSI and HR-MSI inputs, along with the pseudo-color fusion results of different methods using channels (17, 12, 10).

From the visual comparison, classical methods such as CNMF and HySure fail to simultaneously preserve spatial sharpness and spectral consistency, leading to oversmoothed textures or structural artifacts in regions like rooftops and parking lots. DL-based methods like CuCaNet and MIAE enhance spatial details to some degree but tend to introduce color distortions or unnatural textures. Diffusion-based PLRDiff shows improvements in spatial clarity, yet still suffers from notable spectral deviations.

In contrast, our proposed CDPS method achieves a more effective balance between spatial detail restoration and spectral fidelity. It generates sharper edges, finer textures, and smoother color transitions, without introducing visible artifacts. Notably, structural elements such as vehicles and building outlines are well preserved, and the overall color distribution closely matches that of the HR-MSI. These results highlight the strong ability of our method to adaptively integrate spectral and spatial information, benefiting from its subspace-driven modeling. The superior visual quality clearly confirms the effectiveness and robustness of CDPS in addressing real-world HSI-MSI fusion challenges.

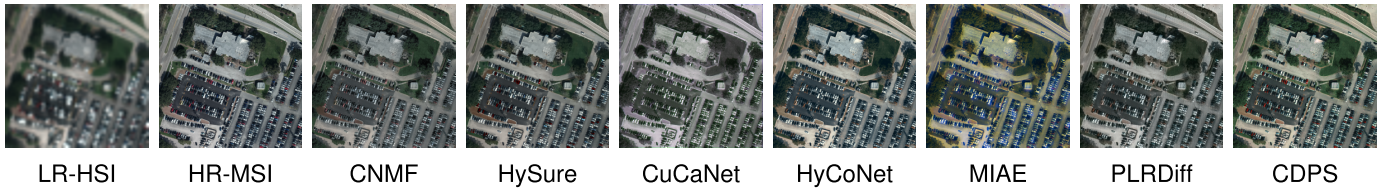


Fig. 10. Visual results obtained by different methods on the Houston dataset.

TABLE IV

ABLATION STUDY ON ABUNDANCE MAPS AND ENDMEMBERS

Methods	PSNR $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	SSIM $\uparrow$	UIQI $\uparrow$
CDPS w/o A	21.73	11.80	13.71	0.282	0.853
CDPS w/o E	39.45	3.33	2.08	0.971	0.994
CDPS	41.95	2.58	1.60	0.979	0.997

E. Ablation Study

1) *Abundance Maps and Endmember Matrix*: As our method achieves excellent fusion performance by simultaneously updating the Abundance Maps and Endmember matrix during the sampling process, we further conduct an ablation study to analyze the individual contributions of these two components. Specifically, we fix either the Abundance Maps or the Endmember matrix as constants during sampling to investigate the impact of updating only one component.

First, we apply an unmixing algorithm to the input HR-MSI to extract the abundance maps, which are then used to replace the corresponding component that would otherwise be updated during sampling. The results of this setting are shown in the first row of Table IV. Similarly, we apply the same unmixing algorithm to the LR-HSI to obtain the endmember matrix, which is then fixed during sampling instead of being updated; the corresponding results are presented in the second row of Table IV.

From the results, it is evident that omitting the update of the abundance maps during sampling leads to a significant performance drop in fusion quality. In contrast, simultaneously updating both components yields the best fusion performance.

2) *Combination of Spatial and Spectral Network Architectures*: Since our proposed method does not rely on any pretrained models but instead employs manually designed lightweight spatial and spectral networks, we conduct a detailed investigation into the architectural configurations of these two networks.

For the UNet-based spatial network, we primarily explore the effect of the `channel_mult` parameter, which controls the number of channels in the feature maps at different encoder layers. For example, if `channel_mult` = (1, 2), the first two layers of the encoder will have 64 and 128 channels, respectively. We analyze four configurations: (1, 1), (1, 2), (1, 2, 4), and (1, 2, 3, 4). For the MLP-based spectral network paired with the spatial network, we focus on the `num_hidden` parameter, which denotes the number of neurons in the hidden layers of the MLP. We investigate three configurations: (256, 256), (256, 512, 256), and (256, 512, 512, 256).

TABLE V

ABLATION STUDY ON COMBINATION OF SPATIAL AND SPECTRAL NETWORK ARCHITECTURES. THE THREE VALUES IN THE TABLE REPRESENT PSNR $\uparrow$ / SAM $\downarrow$ / TIME, RESPECTIVELY

Methods	(256,256)	(256,512,256)	(256,512,512,256)
(1,1)	41.60 / 2.85 / 60	41.66 / 2.93 / 60	41.74 / 2.89 / 60
(1,2)	41.49 / 3.08 / 80	41.95 / 2.58 / 80	42.03 / 2.64 / 80
(1,2,4)	41.17 / 3.04 / 107	41.89 / 2.58 / 107	41.83 / 2.77 / 107
(1,2,3,4)	41.28 / 3.02 / 113	41.96 / 2.55 / 113	41.92 / 2.62 / 113

Table V presents the results of different combinations of spatial and spectral networks. As shown, the configuration with `num_hidden` = (256, 256) in the spectral network generally yields inferior performance in terms of quantitative metrics. Similarly, setting `channel_mult` = (1, 1) for the spatial network also results in poor performance. Moreover, we observe that the size of the spectral network has a negligible impact on the sampling time, while increasing the size of the spatial network tends to degrade the overall performance. Considering both fusion accuracy and computational efficiency, we choose the configuration (1, 2) for the spatial network and (256, 512, 256) for the spectral network in our final model.

F. Analysis of Hyperparameters

1) *Impact of R* : In Eq. (12), we decompose the high-dimensional HR-HSI data $\mathbf{Z}$ into two low-dimensional components. The parameter R is the number of endmembers. In this part, we delve into the impact of R on our experiments. Fig. 11 presents the PSNR results for each dataset with different rank settings. It is observed that the PSNR reaches its peak and stabilizes when R varies between 8 and 10. This phenomenon arises from the fact that when R is too small, the limited number of parameters fails to adequately capture the complexity of the data, while a larger number of endmembers introduces more parameters to optimize, consequently escalating uncertainty during the optimization process. Hence, considering that increasing R also prolongs the running time, we recommend setting it to 8.

2) *Impact of λ_1 and λ_2* : As depicted in Eq. (22) and Eq. (23), the parameters λ_1 and λ_2 are used to balance the two observations. We validate the selection of these two parameters on the Pavia dataset. The initial idea was to set $\lambda_1 = \lambda_2 = \frac{whB}{WHb}$ to ensure equal contribution of the two observation terms in terms of pixel quantity. In this case, we have $\lambda_1 = \lambda_2 = \frac{64 \times 64 \times 103}{128 \times 128 \times 4}$. For convenience, we denote Λ as $\frac{whB}{WHb}$. Subsequently, we conducted experiments around the Λ value to validate our hypothesis. Fig. 12 analyzes the influence

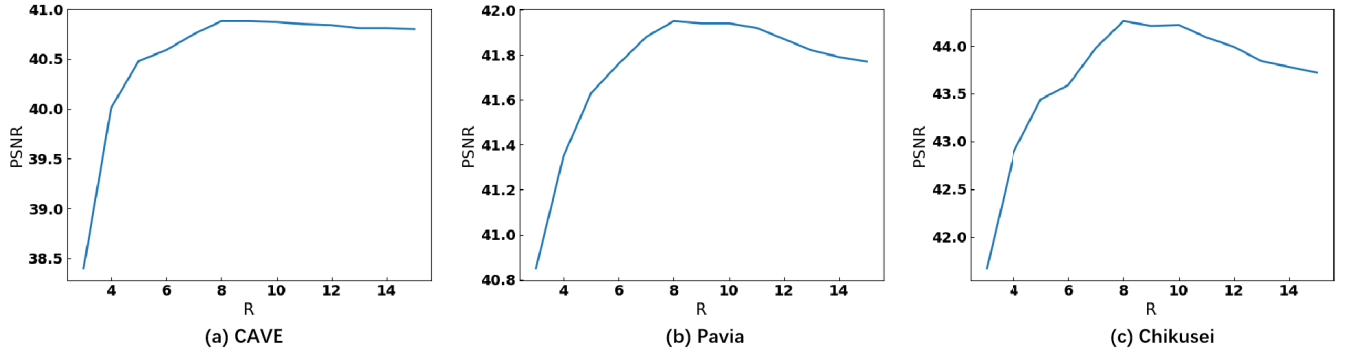
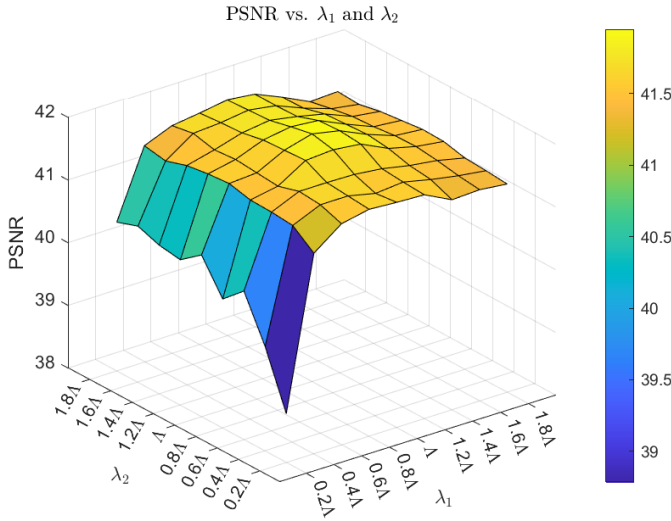


Fig. 11. PSNR results under different rank settings.

Fig. 12. Sensitivity analysis of the parameters λ_1 and λ_2 . $\Lambda = \frac{whB}{WHb}$.

of λ_1 and λ_2 on the PSNR metric, indicating that our method performs better when $\lambda_1 \geq 0.4\Lambda$. Upon closer observation, it can be noticed that the surface exhibits a slight convexity near the (Λ, Λ) region. Therefore, for all experiments, we set $\lambda_1 = \lambda_2 = \Lambda$.

3) *Impact of ρ_1 and ρ_2* : In Eq. (22) and Eq. (23), two additional parameters, ρ_1 and ρ_2 , represent the step size for the reconstructed image to approach LR-HSI and HR-MSI, respectively. These parameters are similar to the learning rates used in training models. As the sampling steps increase, we gradually decrease the values of ρ_1 and ρ_2 , allowing the reconstructed HSI to approach LR-HSI and HR-MSI more rapidly after spatial and spectral degradation. The simple definition for the gradual decay of ρ_1 and ρ_2 is as follows:

$$\begin{aligned} \rho_1 &= P_1 * 0.998^{N-1-t}, \\ \rho_2 &= P_2 * 0.998^{N-1-t}, \end{aligned} \quad (26)$$

where t represents the current sampling step, and P_1 and P_2 represent the initial values of ρ_1 and ρ_2 , which are the values at $t = N - 1$. Next, we explored the setting of the initial values P_1 and P_2 on three datasets. According to Fig. 13, excessively large or small values of P_1 and P_2 do

TABLE VI
FUSION RESULTS OF OUR METHOD WITH INCREASING RANK R

Rank R	PSNR	SAM	Time
10	41.93	2.59	86
20	42.06	2.60	161
30	42.28	2.58	239
40	42.23	2.56	325
50	42.20	2.54	405
60	42.49	2.51	486
70	42.16	2.52	571
80	42.17	2.53	675

not contribute to performance improvement. It is essential to find a moderate and mutually compatible pair of values to achieve better performance. Further analysis across the three datasets shows that performance is generally improved when $P_2 = 0.05$, with optimal results obtained when P_1 is set to 15. Consequently, in the experiments conducted on the three datasets, we consistently set P_1 to 15 and P_2 to 0.05.

G. Comparison With Nonlinear Models

The linear mixture model (LMM) has been widely adopted in hyperspectral unmixing and fusion tasks due to its simplicity and physical interpretability. However, in real-world scenarios involving nonlinear interactions or complex material mixtures, the LMM may fail to fully capture the spectral variability. To investigate this limitation, we present two studies:

First, an experimental study that shows the LMM can achieve promising results. Within the LMM, the number of endmembers (also known as the rank parameter) critically governs reconstruction quality by defining the model's representational capacity. Empirical evidence suggests that higher rank generally enhances representational capacity and we can assume LMM with high rank are close to nonlinear models since the endmembers are abundant. We evaluated fusion quality across ranks 10–80 to determine whether PSNR scales with increased rank. Since increasing the rank R generally requires longer optimization to reach convergence. Therefore, in this experiment, both the spatial and spectral networks are trained for 2000 iterations, and 2000 diffusion steps are used during inference to ensure the fidelity of high-rank subspace representations. The observed saturation in reconstruction

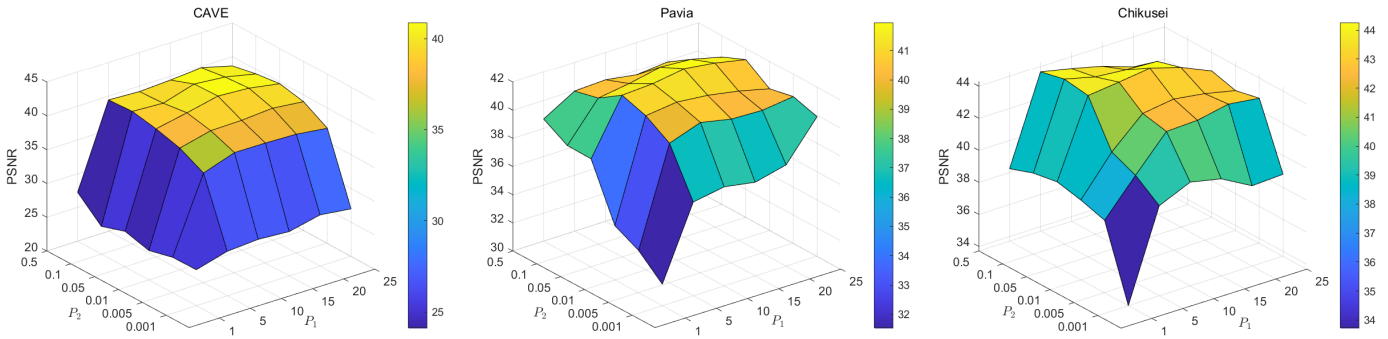


Fig. 13. Sensitivity analysis of the parameters ρ_1 and ρ_2 . P_1 and P_2 represent the initial values of ρ_1 and ρ_2 , respectively.

TABLE VII

COMPARISON WITH A NONLINEAR FUSION MODEL (GBM-CNMF)

Method	Type	PSNR (dB)	SAM
GBM-CNMF [56]	Nonlinear	39.86	2.94
Ours	Linear	42.49	2.51

quality beyond moderate ranks (TABLE VI), along with the stable PSNR achieved by various ranks, suggest that the linear model already achieves near-optimal performance for this task. While nonlinear models can theoretically approximate complex mappings with higher-rank representations, this does not inherently guarantee superior empirical outcomes, as their effectiveness remains contingent on data compatibility and optimization constraints.

Second, comparison with nonlinear model. To further investigate the potential of nonlinearity in fusion tasks, we compare our method with a nonlinear baseline. Due to the limited availability of nonlinear fusion techniques, we select GBM-CNMF [56], a representative method based on nonlinear modeling of endmember interactions. After our tuning, the rank of 30 was found to yield the best performance for this comparative method, and thus we set it accordingly. As shown in TABLE VII, our method significantly outperforms GBM-CNMF. While this suggests that our framework under the LMM assumption is already highly effective, the performance gap may also be attributed to differences in model design and optimization strategies. This comparison motivates future work to explore the integration of nonlinear mixture models within our diffusion-based architecture to better handle even more challenging and complex fusion scenarios.

H. Robustness to Observation Noise

To quantitatively evaluate the robustness of our method under realistic noisy conditions, we conducted experiments on the Pavia University dataset with synthetic Gaussian noise added to both the LR-HSI and HR-MSI inputs. The noise was calibrated to achieve a Signal-to-Noise Ratio (SNR) of 35 dB, simulating challenging but realistic acquisition conditions. All diffusion priors were trained directly on these noisy observations to ensure a fair assessment of noise robustness.

Table VIII presents the quantitative comparison under these noisy conditions. Our proposed CDPS method demonstrates

TABLE VIII

PERFORMANCE COMPARISON ON NOISY PAVIA UNIVERSITY DATASET (INPUT SNR = 35 dB). THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE WITH UNDERLINE

Methods	PSNR $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$	SSIM $\uparrow$	UIQI $\uparrow$
CNMF	39.17	3.07	1.89	<u>0.975</u>	0.995
HySure	37.78	3.19	2.30	0.971	0.993
CuCaNet	37.82	8.86	3.93	0.945	0.977
HyCoNet	38.67	5.79	3.84	0.961	0.984
MIAE	<u>40.99</u>	<u>2.89</u>	<u>1.71</u>	0.973	0.996
PLRDiff	40.36	3.33	1.91	0.972	0.995
S ² CycleDiff	37.31	3.78	2.29	0.970	0.995
CDPS	41.59	2.84	1.68	0.977	0.997

superior performance across all evaluation metrics, achieving the highest PSNR (41.59 dB), lowest SAM (2.84), lowest ERGAS (1.68), and best structural similarity measures (SSIM: 0.977, UIQI: 0.997).

Notably, CDPS outperforms the second-best method (MIAE) by 0.60 dB PSNR while achieving superior spectral preservation. The performance gap is particularly significant against other diffusion-based approaches: PLRDiff and S²CycleDiff show substantial degradation under noise, with PSNR drops of 1.23 dB and 4.28 dB respectively. These results demonstrate the inherent denoising capability of our framework. The iterative posterior sampling balances fidelity to noisy observations with learned data priors, leveraging the noise-reversal dynamics of diffusion models.

I. Computational Efficiency

In all the experiments mentioned above, we conducted them on a device equipped with an Intel Core i7-14700K CPU, GeForce RTX 4090 GPU, and 64GB of memory. Except for the HySure method, which was experimented on the Matlab platform, all other methods were implemented using the PyTorch framework. In this subsection, we primarily consider two factors: model parameters and inference time, to compare the consumption and performance of different methods. Detailed information can be found in Table IX. Traditional methods may have short runtime, but due to limitations inherent in their algorithms, they often fail to achieve a sufficiently high fusion accuracy. In neural network-based subspace methods, the MIAE method has few model parameters and short runtime, whereas other methods, although

TABLE IX
MODEL PARAMETERS AND INFERENCE TIME OF DIFFERENT METHODS ON
THREE DATASETS

Dataset	Method	PSNR	Params(M)	Time(s)
CAVE	CNMF	38.88	×	4
	HySure	39.96	×	3
	CuCaNet	37.13	2.72	156
	HyCoNet	37.11	0.45	79
	MIAE	39.98	0.00129	17
	PLRDiff	35.83	391.05	78
	S ² CycleDiff	38.06	64.40	62
	CDPS	40.88	2.84	76
Pavia	CNMF	39.89	×	7
	HySure	38.01	×	5
	CuCaNet	38.28	2.74	405
	HyCoNet	39.19	0.47	237
	MIAE	41.64	0.00245	19
	PLRDiff	40.04	391.05	79
	S ² CycleDiff	38.04	606.44	296
	CDPS	41.95	3.13	80
Chikusei	CNMF	41.19	×	8
	HySure	41.69	×	5
	CuCaNet	40.06	2.74	431
	HyCoNet	43.84	0.47	263
	MIAE	43.71	0.00257	21
	PLRDiff	40.94	391.05	80
	S ² CycleDiff	40.83	684.63	312
	CDPS	44.26	3.15	82

slightly superior in terms of parameter count, have significantly longer runtimes. For these methods, there is still room for improvement in fusion quality based on the PSNR metric. Our proposed method, as a diffusion model-based approach, holds significant advantages over two other diffusion model-based methods. PLRDiff has over 100 times the parameter count of CDPS, with comparable inference times. Compared to S²CycleDiff, on the CAVE dataset, CDPS has only about 1/30th of the parameter count and a similar runtime. On the Pavia and Chikusei dataset, CDPS has only about 1/200th of the parameter count of S²CycleDiff and a runtime over three times shorter.

V. CONCLUSION

In this paper, we propose a coupled diffusion posterior sampling (CDPS) method for unsupervised HSI and MSI fusion. In the proposed method, the diffusion posterior sampling is leveraged to generate HR-HSI by taking the observed LR-HSI and HR-MSI as conditions. To speed up the training phase and decrease the number of required training samples, we decompose the original HSI into two parts—abundance maps and endmembers. Thus, we can use two simple and low-dimensional diffusion models to generate a high-dimensional HSI. In addition, the simple network allows us to reduce the training samples. This decomposition allows us to train two simple, low-dimensional diffusion models. Crucially, these models are trained in an unsupervised manner, using only the input test image pair as training data. Thus, we propose a highly practical fusion method that operates without any external training datasets or pre-trained models. Experimental results on three datasets show that the proposed CDPS method achieves better fusion results and takes less time. In the future,

we will design fast sampling methods and effective priors of the latent variables in the generation process.

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REFERENCES

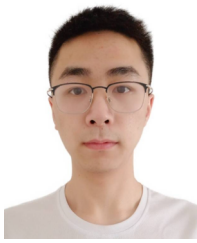
- [1] B. Wang, Y. Xu, Z. Wu, T. Zhan, and Z. Wei, "Spatial-spectral local domain adaption for cross domain few shot hyperspectral images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539515.
- [2] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.
- [3] Y. Xu, Z. Wu, J. Chanussot, P. Comon, and Z. Wei, "Nonlocal coupled tensor CP decomposition for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 348–362, Jan. 2020.
- [4] L. Wang, Z. Xiong, G. Shi, F. Wu, and W. Zeng, "Adaptive non-local sparse representation for dual-camera compressive hyperspectral imaging," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 10, pp. 2104–2111, Oct. 2017.
- [5] G. Vivone, "Multispectral and hyperspectral image fusion in remote sensing: A survey," *Inf. Fusion*, vol. 89, pp. 405–417, Jan. 2023.
- [6] Z. Wang, B. Chen, H. Zhang, and H. Liu, "Unsupervised hyperspectral and multispectral images fusion based on nonlinear variational probabilistic generative model," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 721–735, Feb. 2022.
- [7] X. Tian, K. Li, W. Zhang, Z. Wang, and J. Ma, "Interpretable model-driven deep network for hyperspectral, multispectral, and panchromatic image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 14382–14395, Oct. 2024.
- [8] R. Dian, A. Guo, and S. Li, "Zero-shot hyperspectral sharpening," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12650–12666, Oct. 2023.
- [9] M. Zhou et al., "A general spatial-frequency learning framework for multimodal image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 7, pp. 5281–5298, Jul. 2025.
- [10] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Reciprocal transformer for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102148.
- [11] W. He, Z. Wu, N. Yokoya, and X. Yuan, "An interpretable and flexible fusion prior to boost hyperspectral imaging reconstruction," *Inf. Fusion*, vol. 111, Nov. 2024, Art. no. 102528.
- [12] J. Qu et al., "A principle design of registration-fusion consistency: Toward interpretable deep unregistered hyperspectral image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9648–9662, May 2025.
- [13] W. Wang et al., "Enhanced deep blind hyperspectral image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 3, pp. 1513–1523, Mar. 2023.
- [14] W. Sun et al., "Generating high-resolution hyperspectral time series datasets based on unsupervised spatial-temporal-spectral fusion network incorporating a deep prior," *Inf. Fusion*, vol. 111, Nov. 2024, Art. no. 102499.
- [15] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.
- [16] L. Zhang, J. Nie, W. Wei, Y. Li, and Y. Zhang, "Deep blind hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2388–2400, Jun. 2021.
- [17] Y. Fang, Y. Liu, C.-Y. Chi, Z. Long, and C. Zhu, "CS2DIPs: Unsupervised HSI super-resolution using coupled spatial and spectral DIPs," *IEEE Trans. Image Process.*, vol. 33, pp. 3090–3101, 2024.
- [18] C. Li et al., "CasFormer: Cascaded transformers for fusion-aware computational hyperspectral imaging," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102408.
- [19] S. Jia, Z. Min, and X. Fu, "Multiscale spatial-spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023.

- [20] H. Liu, C. Feng, R. Dian, and S. Li, "SSTF-UNet: Spatial-spectral transformer-based U-Net for high-resolution hyperspectral image acquisition," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 18222–18236, Dec. 2024.
- [21] K. Wang, X. Liao, J. Li, D. Meng, and Y. Wang, "Hyperspectral image super-resolution via knowledge-driven deep unrolling and transformer embedded convolutional recurrent neural network," *IEEE Trans. Image Process.*, vol. 32, pp. 4581–4594, 2023.
- [22] B. Kavar, M. Elad, S. Ermon, and J. Song, "Denoising diffusion restoration models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 23593–23606.
- [23] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," 2020, *arXiv:2011.13456*.
- [24] Y. Wang, J. Yu, and J. Zhang, "Zero-shot image restoration using denoising diffusion null-space model," 2022, *arXiv:2212.00490*.
- [25] H. Chung, J. Kim, M. T. McCann, M. L. Klasky, and J. Chul Ye, "Diffusion posterior sampling for general noisy inverse problems," 2022, *arXiv:2209.14687*.
- [26] Y. Miao, L. Zhang, L. Zhang, and D. Tao, "DDS2M: Self-supervised denoising diffusion spatio-spectral model for hyperspectral image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 12086–12096.
- [27] X. Rui, X. Cao, L. Pang, Z. Zhu, Z. Yue, and D. Meng, "Unsupervised hyperspectral pansharpening via low-rank diffusion model," 2023, *arXiv:2305.10925*.
- [28] Z. Cao, S. Cao, L.-J. Deng, X. Wu, J. Hou, and G. Vivone, "Diffusion model with disentangled modulations for sharpening multispectral and hyperspectral images," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102158.
- [29] W. J. Carper et al., "The use of intensity-hue-saturation transformations for merging SPOT panchromatic and multispectral image data," *Photogrammetric Eng. remote Sens.*, vol. 56, no. 4, pp. 457–467, 1990.
- [30] T.-M. Tu, S.-C. Su, H.-C. Shyu, and P. S. Huang, "A new look at IHS-like image fusion methods," *Inf. Fusion*, vol. 2, no. 3, pp. 177–186, Sep. 2001.
- [31] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.
- [32] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.
- [33] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.
- [34] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.
- [35] K. Zhang, M. Wang, and S. Yang, "Multispectral and hyperspectral image fusion based on group spectral embedding and low-rank factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1363–1371, Mar. 2017.
- [36] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5344–5353.
- [37] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.
- [38] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Hyperspectral computational imaging via collaborative Tucker3 tensor decomposition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 1, pp. 98–111, Jan. 2021.
- [39] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Hyperspectral and multispectral image fusion via deep two-branches convolutional neural network," *Remote Sens.*, vol. 10, no. 5, p. 800, May 2018.
- [40] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial-spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.
- [41] Q. Xie, M. Zhou, Q. Zhao, Z. Xu, and D. Meng, "MHF-Net: An interpretable deep network for multispectral and hyperspectral image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 3, pp. 1457–1473, Mar. 2022.
- [42] W. Wang, W. Zeng, Y. Huang, X. Ding, and J. Paisley, "Deep blind hyperspectral image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4150–4159.
- [43] L. Zhang, J. Nie, W. Wei, and Y. Zhang, "Unsupervised test-time adaptation learning for effective hyperspectral image super-resolution with unknown degeneration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 7, pp. 5008–5025, Jul. 2024.
- [44] S. Li, S. Li, and L. Zhang, "Hyperspectral and panchromatic images fusion based on the dual conditional diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526315.
- [45] J. Qu, J. He, W. Dong, and J. Zhao, "S2CycleDiff: Spatial-spectral-bilateral cycle-diffusion framework for hyperspectral image super-resolution," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 5, pp. 4623–4631.
- [46] F.-A. Croitoru, V. Hondru, R. T. Ionescu, and M. Shah, "Diffusion models in vision: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10850–10869, Sep. 2023.
- [47] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.
- [48] X. Rui, X. Cao, L. Pang, Z. Zhu, Z. Yue, and D. Meng, "Unsupervised hyperspectral pansharpening via low-rank diffusion model," *Inf. Fusion*, vol. 107, Jul. 2024, Art. no. 102325.
- [49] R. H. Yuhas, A. Goetz, and J. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. 3rd Annu. JPL Airborne Geosci. Workshop*, 1992, pp. 147–149.
- [50] L. Alparone, L. Wald, J. Chanussot, C. Thomas, P. Gamba, and L. M. Bruce, "Comparison of pansharpening algorithms: Outcome of the 2006 GRS-S data-fusion contest," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3012–3021, Oct. 2007.
- [51] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [52] Z. Wang and A. C. Bovik, "A universal image quality index," *IEEE Signal Process. Lett.*, vol. 9, no. 3, pp. 81–84, Mar. 2002.
- [53] J. Yao, D. Hong, J. Chanussot, D. Meng, X. X. Zhu, and Z. Xu, "Cross-attention in coupled unmixing nets for unsupervised hyperspectral super-resolution," in *Proc. ECCV*, 2020, pp. 208–224.
- [54] K. Zheng et al., "Coupled convolutional neural network with adaptive response function learning for unsupervised hyperspectral super resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2487–2502, Mar. 2021.
- [55] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522412.
- [56] N. Yokoya, J. Chanussot, and A. Iwasaki, "Hyperspectral and multispectral data fusion based on nonlinear unmixing," in *Proc. 4th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2012, pp. 1–4.



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